

Unit Conversions Practice

- There are 5280 feet in one mile
- There are 3.28 feet in one meter
- There are 1.609 kilometers in one mile
- There are 0.224 lbs in one Newton
- There are 0.454 kg in one pound
- There are 73 gallons in 2 barrels
- There are 1.05 quarts in one liter
- There are 4 quarts in one gallon

Do the following SI conversions, using your knowledge about prefixes:

- 1) How many pm are in 2.38×10^{-2} km?

$$\frac{2.38 \times 10^{-2} \text{ km}}{1} \times \frac{10^3 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ pm}}{10^{-12} \text{ m}} = 2.38 \times 10^{13} \text{ pm}$$

Do the following one-step unit conversions:

- 2) Convert 56 miles to feet.

$$295680 \text{ ft}$$

- 3) Convert 1350 kilograms to pounds.

$$\frac{1350 \text{ kg}}{0.454 \text{ kg}} \times 1 \text{ lb} = 2970 \text{ lb}$$

Do the following multi-step unit conversions:

- 4) Convert 18 barrels to liters.

$$\frac{18 \text{ barrels}}{2} \times \frac{73 \text{ gallons}}{1 \text{ barrel}} \times \frac{4 \text{ quart}}{1 \text{ gallon}} \times \frac{1 \text{ liter}}{1.05 \text{ quart}} = 2503 \text{ L}$$

- 5) Find your age in seconds.

$$\frac{\text{year}}{1} \times \frac{365 \text{ days}}{1 \text{ year}} \times \frac{24 \text{ hr}}{1 \text{ days}} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}}$$

- 6) What is the length of a canoe in μm ? Note that the average canoe is 16.5 feet long.

$$\frac{16.5 \text{ ft}}{3.28 \text{ ft}} \times \frac{1 \text{ m}}{10^6 \mu\text{m}} = 5.03 \times 10^6 \mu\text{m}$$

Do the following multi-step unit conversions with squared/cubed dimensions:

- 7) The density of mercury is $13,579 \text{ kg/m}^3$. What is its density in g/cm^3 ?

$$\frac{13579 \text{ kg}}{\text{m}^3} \times \frac{10^3 \text{ g}}{1 \text{ kg}} \times \frac{10^{-2} \text{ m}}{1 \text{ cm}} \times \frac{10^{-2} \text{ m}}{1 \text{ cm}} \times \frac{10^{-2} \text{ m}}{1 \text{ cm}} = 13.579 \text{ g/cm}^3$$

- 8) You pump up your bicycle tire to 80 psi. What is the pressure in kN/m^2 ?

$$\frac{80 \text{ lbs}}{\text{in}^2} \times \frac{1 \text{ N}}{0.224 \text{ lbs}} \times \frac{1 \text{ kN}}{10^3 \text{ N}} \times \frac{12 \text{ in}}{1 \text{ ft}} \times \frac{12 \text{ in}}{1 \text{ ft}} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 553.3 \text{ kN/m}^2$$